Future first System

**Software Requirement Specification (SRS)**

**Project Name: FutureFirst System**

**Company: THREE D Nepal, From Nepal**

**Development Company: AIONINNO Technologies Private Limited, From India**

**User Roles:**

1. **SuperAdmin (Three D Nepal)**
2. **Community Admin** (Report and Announcement only, this is used by municipality or certain government office of PARTICULAR region)
3. **School Admin/Board** (managing **one** or **multiple** schools under a group or name)
4. **School Administration** (one specific school)
5. **Teacher**
6. **Student/Parent**
7. **External user(optional feature, may be included in next version)**
    a. Normal user means who may **not be currently** enrolled in any education institute. The purpose of adding this feature is to have mass download of our app.
    b. This user will be able to view our course content
    c. View article, video lesson of different topics like government exam, IT and Coding etc.

## User Role Hierarchy (Logical Structure)

1. **SuperAdmin** (Three D Nepal) - **Highest Authority**
2. **Community Admin** (Regional Government/Municipality)
3. **School Admin/Board** (Manages Multiple Schools under a Group)
4. **School Administration** (Manages One Specific School)
5. **Teacher**
6. **Student/Parent - Lowest authority**

**Platform:**
Web, Mobile App(Cross Platform)

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# Student Registration Process

1. Register/Login with third party system
    a. Google, Facebook, Tiktok
2. First time registration of student
    - Student can select interest at first register (similar to tiktok ui)

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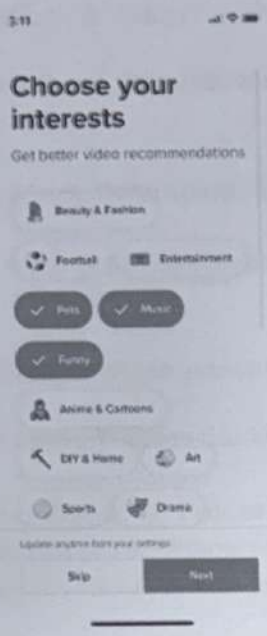


# Features by User Role

## Student Features 🎓

Student Lifetime Unique Serial Number

- Unique ID is generated upon registration
- The same ID is used if a student transfers to another school.
- Previous schools do not have access to new academic data
- But the new school may have access to the academic data/reports of newly enrolled students.

### 1. Homework & Assignment Management

- **Submit & Track Assignments:** Students can submit assignments, track due dates, and receive feedback.
- **Timely Submission Rewards:** Virtual badges for timely submissions (e.g., "Assignment Ninja").
- **Leaderboard & Streaks:** Ranks based on assignment scores and timely submissions, motivating consistent performance.
- **Milestone Badges:** Earn "Gold," "Silver," or other badges upon crossing assignment completion milestones.

### 2. Academic Performance & Reports

- **View Exam Results & History:** Access past exam scores, trends, and detailed performance breakdowns.
- **Real-Time Performance Insights:** Compare percentile rankings within the class/section.
- **Monthly Performance Reports:** Automated notifications summarizing student progress (customizable by the school admin).- notification schedule/notification system set by school admin, not by teacher.

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### 3. Smart Notifications & Exam Reminders

- **Automated Exam Countdown:** Example: "30 days until your exam! How is your preparation going?"
- **Targeted Study Reminders:** More frequent reminders for students in "Poor Performing" groups.
- **Custom Teacher/Admin Messages:** Personalized messages to guide and support students.

### 4. Attendance & Leave Management

- **Request Leave Online:** Students can submit leave requests digitally for approval.
- **Automated Approval Process:** Teachers/Admins can accept or reject requests with a single tap. - First review(approve/disapprove) priority goes to the teacher and second review(approve/disapprove) priority goes to the admin.

### 5. Time Spent Coach (Parental & Admin Monitoring)

- **Track App Usage:** Records time spent on different apps for better study habits.
- **Customizable Settings:** Schools can enable/disable monitoring per class range or school-wide.
- **Data Insights:** Viewable by: SuperAdmins, school admins, and parents & class teacher, same user.

### 6. Gamification & Engagement

- **Virtual Mascot Companion:** Like Duolingo's bird, an engaging character keeps students motivated.
- **Interactive Achievements:** Encourages learning habits through streaks, rewards, and badges.

## Teacher Features

### Academic Management

- Create subject syllabus
- Perform Attendance of Student
- Create, Assign & Review Assignments
- Rate Assignments (Out of 5 Stars) & Provide Custom Feedback
    - Predefined Suggestions: "Improve handwriting", "Fix spelling", "Lack topic understanding" or Custom Message, custom message is **saved** for next time.
- Conduct Online Exams (Quizzes)
    - Import Quiz Questions from PDF/Docs
    - Set Quiz Time & Points for Each Question
    - Schedule Quiz
    - Quiz Types: Open Book (Research Allowed) / Closed Book (Cheating Prevention Enabled)
    - Track App/Tab Switching During Closed-Book Quizzes (Log Switch Events with Duration for Teachers/Admins)
- View Student Academic Profiles & Past Performance
- Communicate with Students (Individual or Group)

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- **Custom Student Groups:** Teachers can create invisible groups (e.g., "Needs Improvement," "Top Performers").
    - **Targeted Alerts & Coaching:** Personalized guidance based on performance levels.

**Performance & Evaluation**

- Exam & Result Calculator
    - Generate Overall Exam Reports (Pass Rate, Class/Section Average, Highest/Lowest Marks, Most Improved Student)
- View & Comment on Announced/Draft Class Routine (Viewable by Department Head/Admin)
- Quick Review of Timetable: "All Good", "Time Conflict" & Custom Comment
- Viewing your own teaching schedule (self-routine only), one teacher **cannot see** the schedule of **another teacher**.

**Administrative & Resource Management (Teacher feature)**

- Apply for Leave (Absent/Present System)
- Upload Study Materials
- Create Subject Syllabus & Weekly Lesson Plan
- Embed YouTube Videos (Playable Within App)
- Upload & Share PDF, Docs, Images, Videos (Client-Side Compression for Large Files)
- Access Previous Teacher's Resources/Assignments (For Same Class/Subject)
    - Current teachers can customize whether a new teacher can access his/her resources.
- View Attendance Reports
    - Number of School Open Days
    - Present Days & Attendance Ratio
- Receive Anonymous Student Feedback (Visible Only to Admin/Teacher)
- Self-Performance Report (Timely Assignment Checking, Student Improvement Ratio, etc.)
- Student Performance Alerts (Identifies Poor-Performing Students)

## Parent Features 👪

- View Student Report
- View Student Result History
- Fee Payment System & Payment History
- View Due Payments & Notifications (Customizable by Admin)
- Receive Announcements (Festive Programs, Events, etc.)
- Comment on Student's Progress (Visible to Teacher/Admin)
- Receive Monthly/Weekly Progress Reports
- Track bus GPS Route

## School Admin/Board🏫👨‍💼, 🏫👨‍💼

Real users may include principal, share holder, investment company etc.

- Each **School** functions autonomously, managing its own administration and operations. However, User **X**, if assigned to multiple schools, can access and oversee aggregated data and management features across all their associated schools.
- **Manages One or Multiple Schools** under a single group or institution name.
- **Access to Aggregate Data** from all schools under their management.
- **Multi-School Announcements** can be sent across all assigned schools.
- **Has Full Access** to all features available to individual **School Administrations** within their group.
- **Financial Performance Tracking**: Shareholders, investment companies, and other stakeholders can access financial reports of schools under their management. This includes revenue from fees, expenses, profit/loss statements, and outstanding dues, providing insights into the financial health of each school or school group.

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- **School Group Assignment**: Only **SuperAdmin (Three D Nepal)** can assign schools to a specific group.

## School Administration Features🏫📋

### Management Systems

- School Calendar Management
- Student Management System
- Exam Result System.
- Faculty/Department Management
  - **Multiple Departments**: The college consists of various departments such as BBA, BCA, and BHM, Computer Engineering etc. Each school can categorize and group departments using custom tags.

**Teacher Management System:**

  - **Teacher Allocation**: A teacher can be assigned to multiple departments based on their expertise.
  - **Hiring Process**: Teachers are hired by individual departments, not by the principal.
  - **Timetable Preparation**: Each department prepares the timetable for its teachers.
  - **Principal's Approval**: The prepared timetable must be reviewed and approved by the principal.
  - **Time Slot Restrictions**:
    - If a time slot is already assigned to a teacher for a class in one department, the same teacher cannot be scheduled for another department at the same time.
    - This ensures no scheduling conflicts across departments.
  - **Timetable Validation**: Proper validation must be implemented to prevent overlapping schedules for teachers across different departments.
- Staff Management System
- Leave Management System
- Payroll Management & HR System
- Library Management System
- Fee Management System
- Financial Accounting System and Reporting
- SMS & Announcement Management
  - Enable/Disable Notifications for Published Results (For Students/Parents)
- Subscription Report
  - How many student cleared subscription fee, dues etc
- ID Card, Certificate, Transcript generation for teacher/student or X user.
- GPS Tracking System.

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### Advanced Analytics & Customization

- Flexible reporting and analysis
  - Identify N Poor-Performing Students (E.g., Bottom 10 Students in Grade 10)
  - Detect Students Failing the Same Subject Multiple Times

- Create & Manage Clubs (Assign Presidents, Set Involved Students)
  - Club EDIT access is given to X user by admin.
- School Customization Options
  - Upload School Logo & Set Theme Colors
  - Select Class Themes (Predefined: Kids-Friendly, Standard, Science, Computer Science, or Custom Themes)

    Figure: "**Choose Interest**" Inspiration of Tiktok UI

  - Preloaded Theme Templates for Schools
- **Accessibility Settings:** Adjust Font Size (Admin/SuperAdmin Can Modify Settings for Specific Parents)

## CommunityAdmin Feature ⚖️: Report and announcement only

Community refers to a city or region government body/office.

- **Umbrella School-Group Data** – View aggregated reports for multiple schools operating under a common name or management.
- **School Report Card** – View detailed performance reports of one or many schools.
- **Attendance Reports** – Track attendance of students and teachers.
- **Fee Collection Report** – View school/school-group-wise fee collection data.
- **Exam & Performance Report** – Analyze student results based on pass/fail rates, with breakdowns by school, subject, and class.
- **Filtered & Aggregate Reports** – Generate detailed reports using custom filters such as school, subject, grade, gender, caste, religion, class range, or faculty.

## SuperAdmin Features 👑 – Centralized Control & Insights

### 1. User & Role Management

- Manage All Users: Create, edit, and deactivate Schools, School Admins, Teachers, Students, and Parents.
- Role-Based Access Control: Assign or modify permissions for different user roles.
- Multi-School Management: Oversee multiple/group of schools/colleges under Three D Nepal.

### 2. School & Faculty Management

- Approve & Onboard New Schools: Manage new school registrations and onboard administrators.
- Customize School Features: Enable/disable specific modules for individual schools.
- Set Academic Structures: Define class levels, grading systems, and academic calendars.

### 3. Performance Monitoring & Analytics

- Student Performance Reports: View trends, percentile rankings, and top/bottom performers.
- Exam & Assignment Insights: Track pass rates, assignment completion rates, and streaks.
- Teacher Effectiveness Reports: Monitor timely assignment reviews, student improvements, and feedback quality.
- Time Spent Analytics: Aggregate app usage data across schools, cities, and regions.
  - Role based usages reporting
  - Usages report filter system

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### 5. Communication & Notifications

- System-Wide Announcements: Broadcast important updates to all schools, admins, teachers, and students, announcements can be targeted to specific group/region/gender or user role.
- Automated Alerts: Set event-based notifications for exams, fee dues, holidays, and academic results.
- Custom Messaging System: Send personalized messages to specific schools, faculty, or students.
- Quick announcement(save announcement template and user ahead), just release it when required on one/two clicks.

**6. Data & Security Management**

- Data Backup & Restore: Ensure periodic backups and restore options for critical school data.
- Privacy & Access Logs: Monitor login activities, data access patterns, and potential security breaches.
- Multi-Factor Authentication: May enforce enhanced security measures for admin logins.
    - School based customization.

**7. System Customization & Settings**

- Enable/Disable Global Features: Toggle features like gamification, notifications, or monitoring tools.
- Multi-Language Support: Set default languages and allow schools to configure either English or Nepali.
- Accessibility Controls: Adjust system-wide settings for font sizes, themes, and user preferences.
    - Default setting or specific for a certain school or group of school/college.

**8. Financial & Subscription Management**

- Manage School Subscriptions: Set pricing plans, renewals, and feature access per school
- Fee Collection & Payment Processing: Monitor payments, track due fees, and send automated reminders.

Financial Reports: Generate revenue insights, school-wise earnings, and expense tracking.

- **SuperAdmin Feature: Manage School Subscriptions & Pricing**

**1. Subscription-Based Revenue Model**

- Per-Student Pricing: Schools are charged based on the number of enrolled students.
- Base Rate: Default pricing (e.g., Rs. 450 per student) with full access to core features.
- Custom Pricing per School: Different schools can have different per-student rates or based on class range, validity is for 365 days or for one academic season.

**2. Payment & Billing System**

- Multi-Payment Methods: Allow online payments via bank transfer, card, or digital wallets(Esewa and Khalti of Nepal)
- Payment Tracking: Monitor payments, due amounts, and upcoming renewals
- Renewal Alerts:
    - Automated reminders for subscription renewals and pending payments.
    - Alerts are sent to schools, parents, and students.
    - Three D Nepal should be notified when a school's subscription is nearing renewal (an alert is triggered 90 days before expiration). After this, the follow-up process begins, including discussions on new features and updates.

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### 3. Admin Dashboard for Revenue & Growth Insights

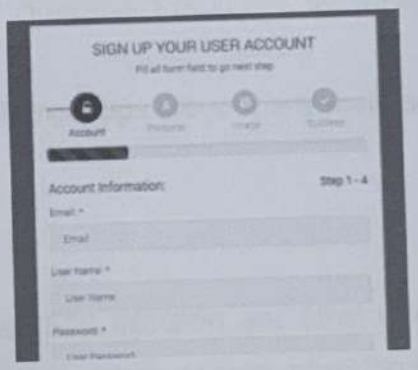
- Revenue Reports: Track total earnings, school-wise payments, and overdue amounts
  - Example user question: How many paid, how many has pending fee, which class is pending.
- Student-Based Revenue Metrics: Analyze payment per student, per school, or per region(average)
- Forecasting & Growth Analytics: Predict revenue trends based on student enrollment data.

## 5. Global Features

- **Offline Mode:** View Data Offline & Sync When Online
- **Calendar Customization:** Choose Nepali or English Calendar (Set by School Admin During Registration)
- **Emergency Alert System:** Notify Users About Natural Disasters (Floods, Earthquakes, etc.) - SMS, MAIL, WHATSAPP, APP
- **Multi-Language Support:** Select Preferred System Language (English, Nepali, etc .)
  - For users: parent/student **only**.

## UI/UX Considerations

- Step-wise forms for data input (Next/Previous navigation). Below is an example of multi-step form:




- Auto-save feature to store input data before submission.
- Users can resume form filling if the app is closed.
- Compatibility with commonly used devices in Nepal.
- Optimize website speed
  - Role-Based Asset Loading – Load only the necessary CSS, JS, and other assets based on the user's role.

## Non-Functional Requirements

### 1. Code Maintainability

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- **Well-Documented and Structured Codebase:** Ensure clear, consistent documentation across all code files, making it easy for developers to understand and maintain.

- **Database Schema and API Documentation:**
  - Provide a comprehensive list of APIs, including:
    - Endpoint URLs
    - Expected input and output data formats (JSON structure)
    - HTTP methods (GET, POST, PUT, DELETE)
    - Authentication and permission requirements

- **Frontend Documentation:**
  - Detail state management strategies
  - Explain the file structure and module organization.
  - Provide usage guidelines for key UI components.

- **App/Web File Code Documentation:**
  - Include **inline comments and function-level documentation**.
  - Maintain a structured guide for major modules, utility functions, and configuration files.
  - Ensure versioning and change logs are properly maintained.

**2. Security Measures**

- Protection against DDoS attacks, SQL injection, and other threats.
- Two-Factor Authentication (2FA) for secure access, ensuring it does not impact usability for students. If 2FA reduces ease of access, it should be optional or excluded for student accounts.
- Real-time database backup.
- User data privacy and compliance with Nepali education regulations.
- **Audit Logs:** Store user device data, IP address, and location.

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# Insight: How animated character is integrated on Duolingo app

**Figure: Real example from App**

## 1. Core Animated Character – Duo the Owl

- **Mascot Role:** Duo, the green owl, is the app's primary mascot, serving as a friendly coach.
- **Emotional Reactions:** Duo changes expressions based on user behavior—cheering for success, looking sad when users miss lessons, or sending reminders.
- **Push Notifications:** Personalized, playful reminders (e.g., "Duo is sad you missed your lesson!").

## 2. Interactive Animations

- **Encouragement & Feedback:** The character reacts in real-time when users complete lessons, make mistakes, or break streaks.
- **Animations for Streaks & Achievements:** Celebratory effects (e.g., confetti, dancing Duo) when users reach milestones.
- **Failure & Retry Encouragement:** Expressive animations encourage users to keep going despite mistakes.

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